

(a) Existing Methods: Key Limitations

RRT*

Explores uniformly
→ many wasted samples



Informed RRT*

Euclidean ellipse only
→ loose focus region



BIT*

Euclidean heuristics
→ mis-ranked edges



All of the above

Ignore collision data
→ repeat mistakes



(b) RIT* + CARM: Our Solutions

Riemannian Informed Set

Tighter focus
→ 60% fewer wasted samples

Whitened Sampling

Handles anisotropy
→ rejection-free sampling

Cascading Edge Evaluation

Riemannian heuristics
→ 3-level lazy filtering

CARM

Collisions → cost map
→ learns obstacle layout